Data Quality Assessment –Group 1
ECG-Based Indicators of Underlying Cardiac Disease and Arrhythmia Presence

Git Hub Repository: https://github.com/Tarra242/Data-Wrangling-Group-1-
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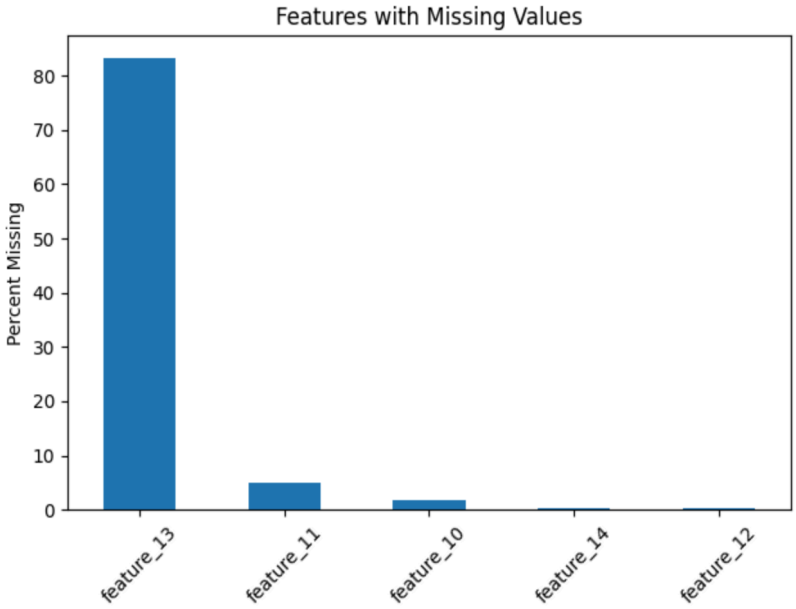
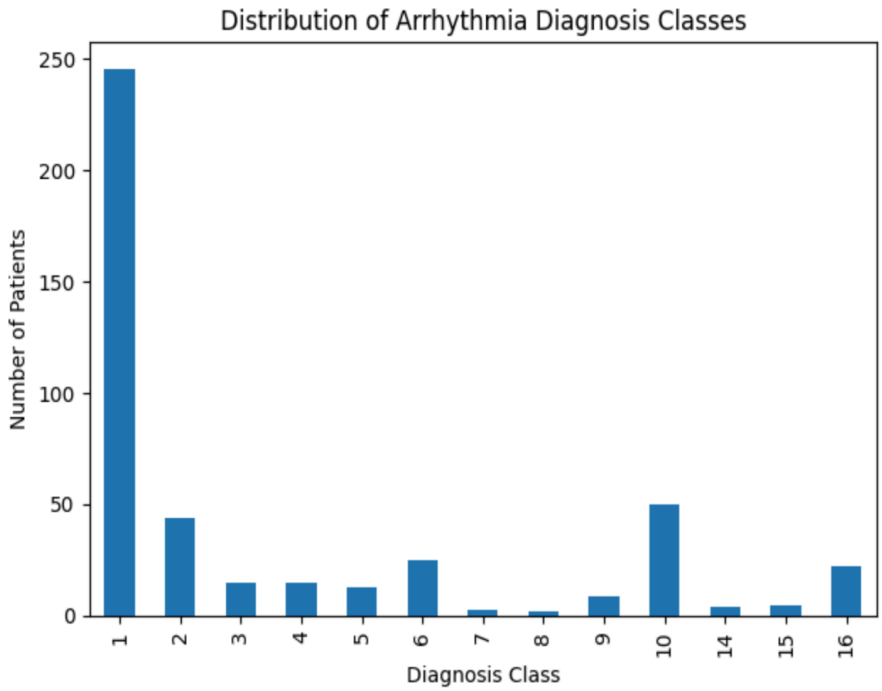
This data quality assessment evaluates the Arrhythmia dataset obtained from the UCI Machine Learning Repository (Guvenir et al., 1997).The dataset contains 452 patient records with 279 electrocardiogram (ECG) features and a cardiologist assigned diagnosis label representing arrhythmia classification. The variables include demographic descriptions (e.g., age and sex) alongside multi dimensional ECG quantitative measurements from standard 12 lead electrocardiograms.

The purpose of this assessment is to determine whether the dataset is suitable for proper analysis in examining ECG difference between a normal rhythm and arrhythmia subtypes. Given the large number of features and the presence of missing values, a thorough evaluation is necessary prior to visualization or modeling. Guided by the data quality framework proposed by Weiskopf and Weng (2013), this assessment examines completeness, plausibility, concordance and currency across three domains; demographics, ECG features and diagnosis labels. In addition, descriptive statistics are included to summarize distribution, demographic characteristics and selected ECG measurements in order to characterize structure and variability of the dataset.

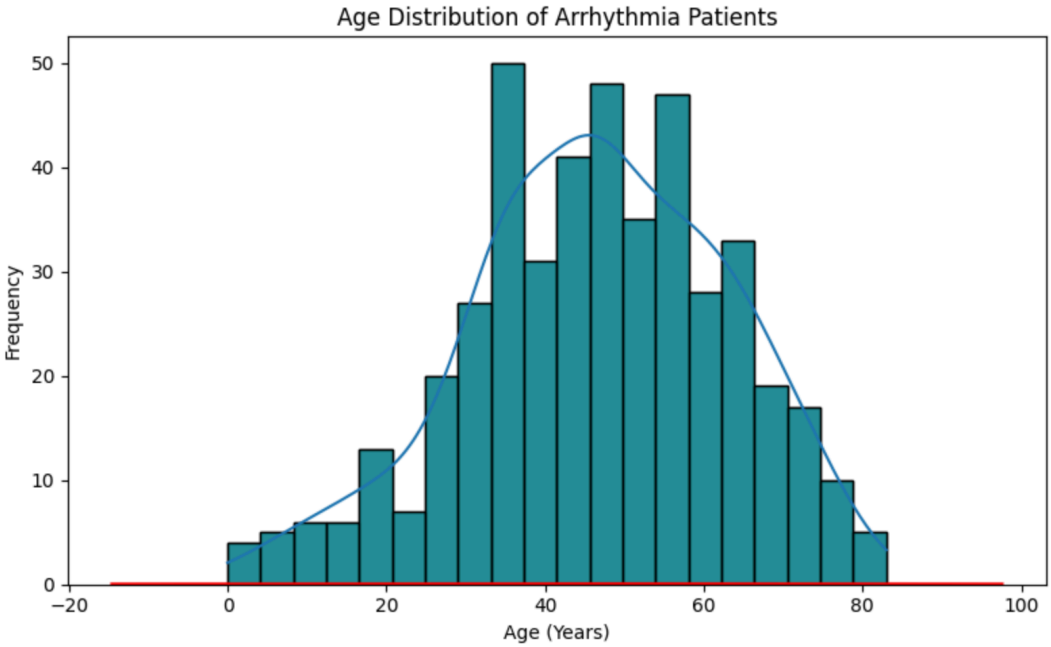


Completeness was assessed by calculating the percentage of missing values for each variable. Missing variables were originally represented as “?” then converted to NaN prior to analysis. Overall, the dataset demonstrates high completeness among most variables. (See bar chart.) The majority of ECG features contain no missing data.

The diagnosis distribution contains 16 classes representing normal cardiovascular rhythms and various arrhythmia subtypes. Class 1 (normal ECG) constitutes 54.2% of the dataset and represents the majority. The second most frequent arrhythmia class includes Class 10 (11.1%), and Class 2 (9.7%) and the remaining subclasses account for fewer than 5% of observations and a small amount constitute less than 1%. This distribution demonstrates a class imbalance, particularly among rarer arrhythmia subtypes. This imbalance may influence downstream classification performance (He & Garcia, 2009) and should be considered when interpreting the data set and corresponding results.



Demographic characteristics were analyzed and “age” was summarized using descriptive statistics. The mean age of people studied in this dataset was 46.5 years old, with an interquartile range of 36-58 years and a maximum age of 83 years old. The age distribution reflects a broad representation of adult patients. A minimum value of 0 years



(representing newborn patients) was observed and will be evaluated further during plausibility assessment. Sex was coded as a binary variable (0 = female, 1 = male). The division of the two was divided well, with 55.1% of patients representing as male and 44.9% representing as female. This demonstrates balanced comparisons without significant imbalance.

Beyond demographic variables, the dataset contains 275 ECG measurements representing temporal intervals, waveform durations, amplitudes and morphological characteristics extracted from the standard 12 lead ECG's. These features include timing intervals such as QRS duration, PR and QT intervals (Kligfield et al., 2007), as well as amplitude measurements. Together, these variables show electrical conduction and waveform patterns that are clinically relevant for identifying arrhythmia subtypes.

Plausibility of ECG variables was assessed by reviewing summary statistics and distributional ranges across a subset of features. The ECG measurements are formatted as continuous numeric variables and show reasonable variability. While individual variables exhibit variability and include both positive and negative waveform amplitudes, no widespread structural abnormalities or significant outliers were observed across the dataset. Overall, the ECG feature set appears structurally sound and physiologically plausible for further analysis.

Each of the 452 records contains a complete set of ECG measurements aligned with a corresponding diagnosis label. No duplicated observations were seen, and no missing diagnostic classifications were detected. The dataset is organized as a single patient level table, minimizing discrepancies and supporting alignment between predictors and outcome variables.

Overall, the dataset demonstrates strong structural completeness, plausibility, and concordance domains. Missing data is minimal for the majority of features, demographic distributions are reasonable, and ECG measurements exhibit coherent numerical formatting. While outlier values and class imbalances are present, these do not compromise the overall usability of the dataset. Collectively, the Arrhythmia dataset appears well organized and provides a well organized reference for exploratory analysis of ECG features across arrhythmia subtypes.

References

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